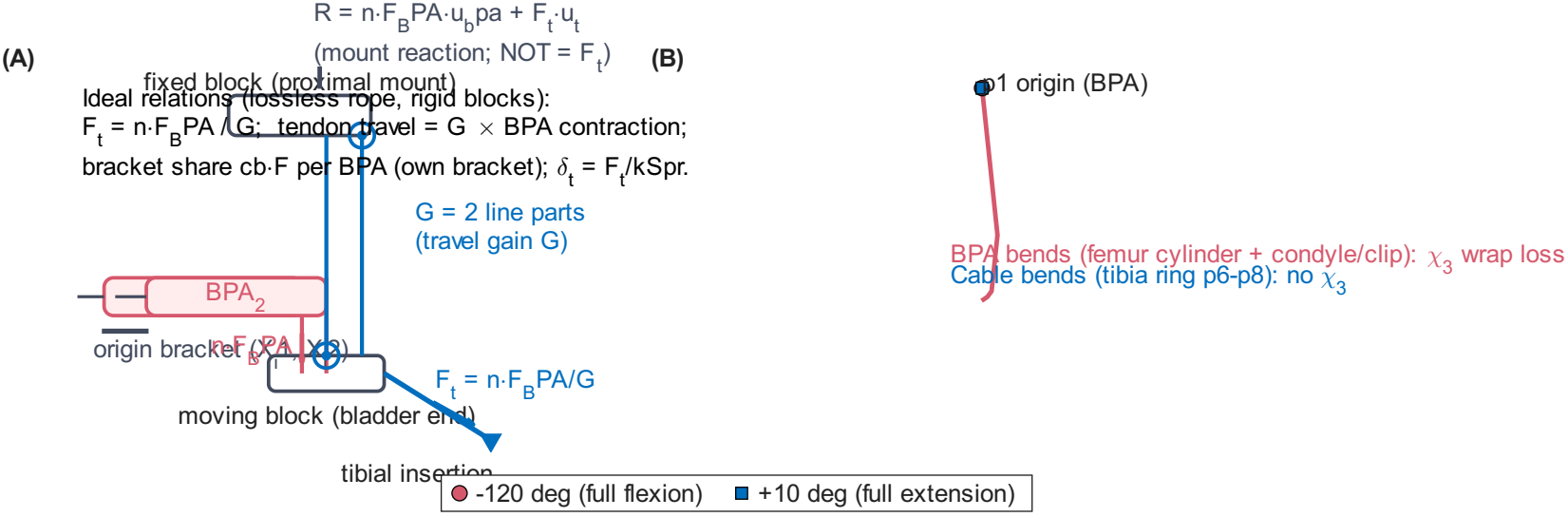
(A)
R = n·F_B PA·u_b pa + F_t·u_t
(mount reaction; NOT = F_t)
(B)
fixed block (proximal mount)
Ideal relations (lossless rope, rigid blocks):
F_t = n·F_B PA / G;  tendon travel = G × BPA contraction;
bracket share cb·F per BPA (own bracket); δ_t = F_t/kSpr.
G = 2 line parts
(travel gain G)
BPA_2
origin bracket (XA, X3)
n·F_B PA (XA, X3)
F_t = n·F_B PA/G
moving block (bladder end)
tibial insertion
p1 origin (BPA)
BPA bends (femur cylinder + condyle/clip): χ_3 wrap loss
Cable bends (tibia ring p6-p8): no χ_3
-120 deg (full flexion)
+10 deg (full extension)